

Manohar Mahadev Athani

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Professional Summary

Recently graduated ECE engineer from BIT Bangalore with strong foundations in DSA, backend development, Low-Level Design, and High-Level Design. Built production-grade Node.js microservices deployed on Google Cloud Platform (GCP), applied SOLID principles across real-world LLD projects, and designed scalable distributed architectures covering load balancing, caching, asynchronous processing, and cloud-native deployments.

Education

- Bangalore Institute of Technology** Bangalore, India
B.E. in Electronics and Communication | CGPA: 8.83 May 2026
- SRA Composite PU College** Banahatti, India
PUC | Percentage: 84% 2022

Experience

- Mindmatrix** Remote
Android App Development Intern — Generative AI 2026
- Built Android application features using **Kotlin** and **Jetpack Compose**, implementing clean UI components with **MVVM architecture** for clear separation of concerns.
- Integrated **REST APIs** for real-time data fetching and managed local data persistence using **Room Database**, ensuring offline-first functionality.
- Implemented dynamic list rendering with **RecyclerView** and optimized app performance through efficient state management in the MVVM pattern.

Projects

- Notification Service** | *Node.js, Express.js, BullMQ, Redis, PostgreSQL, Socket.IO, JWT, Docker, GCP*
- Production-grade **multi-channel notification microservice** supporting **Email, SMS, and In-App** delivery via a unified **REST API**; used **BullMQ + Redis** for **async job processing** and **Socket.IO** for real-time delivery with **JWT authentication**.
- Deployed on **Google Cloud Run** with **Cloud SQL (PostgreSQL)**; containerized with **Docker** enabling **horizontal scaling** in **cloud-native**, microservice-based architectures.
- URL Shortener Service** | *Node.js, Express, Redis, PostgreSQL, Docker*
- High-throughput REST API** handling **10,000+ redirects/day** with **sub-50ms latency** using **Base62 encoding** and **Redis caching** achieving **95% cache hit rate**.
- Containerized with **Docker**; designed scalable **HLD** with separate tiers **API gateway**, app server, cache, and DB enabling **horizontal scaling**.
- BookMyShow Clone** | *C++, LLD, OOP, SOLID, Design Patterns*
- Ticket booking system in **C++** applying all **5 SOLID principles** across **15+ classes** with **Singleton, Factory, and Strategy** patterns for pluggable payment and theatre management.
- Implemented **time-bound seat locking** for **concurrent conflict handling**; modeled full domain with **UML class diagrams** following **Separation of Concerns**.

Technical Skills

- Languages:** C, C++, JavaScript
- Backend & Web:** Node.js, Express.js, REST APIs, Microservices, Asynchronous Processing, WebSockets, HTML, CSS
- Databases & Caching:** PostgreSQL, MongoDB, Redis, SQL
- Message Queues:** BullMQ, Redis Queues, Event-Driven Architecture
- Cloud & DevOps:** Google Cloud Platform (GCP), Cloud Run, Cloud SQL, Docker, Containerization, CI/CD
- System Design:** Class Design, UML Diagrams, Design Patterns (Singleton, Factory, Strategy), Microservices, Load Balancing, Caching, DB Sharding, API Gateway, CAP Theorem, CDN
- Core Concepts:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks
- Principles & Tools:** SOLID, DRY, KISS, Git, Docker, VS Code

Extracurricular Activities

- Active member of the **Robotics Club**, Bangalore Institute of Technology.
- Participated in **Inter-college Coding Hackathon (2023)**.